

- 5 (a) A stationary electron is placed in a field of force, and it experiences a force which is **same** in direction as the field lines. For each of the cases below, state and explain whether the field can be

(i) magnetic,

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 [1]

(ii) electric,

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 [1]

(iii) gravitational.

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 [1]

- (b) (i) Calculate the magnitude of the electric field strength required to maintain an electron in a fixed position above the ground in the gravitational field of the Earth near its surface.

electric field strength = N C^{-1} [2]

- (ii) Hence explain why gravitational effects are ignored when considering the motion of electrons in electric fields.

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 [1]

- (c) An electron is injected at a speed $v = 7.0 \times 10^6 \text{ m s}^{-1}$ at an angle θ into a uniform horizontal magnetic field of flux density $3.14 \times 10^{-5} \text{ T}$. It moves in a helical path as shown in Fig. 5.1.

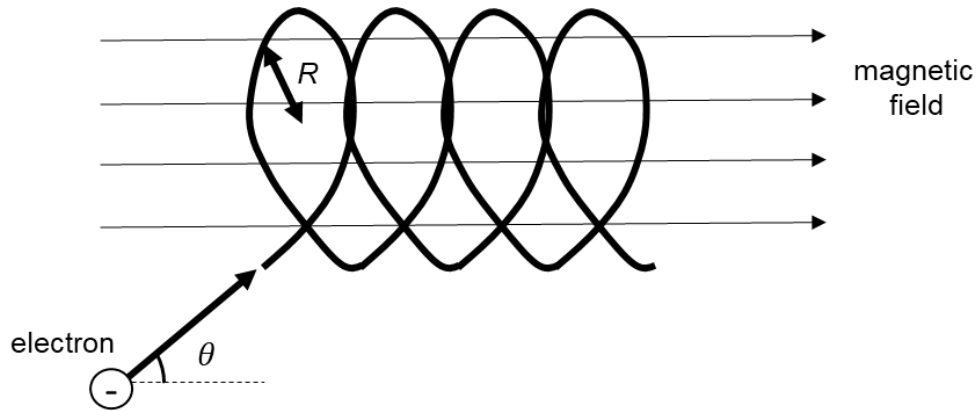


Fig 5.1

- (i) Explain the motions attributed to the vertical and horizontal components of the electron's velocity which lead to its helical path.

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..... [2]

- (ii) Assuming that $\sin \theta = 3/5$ and $\cos \theta = 4/5$, show that the velocity component of the electron perpendicular to the magnetic field is $4.2 \times 10^6 \text{ m s}^{-1}$.

[1]

- (iii) Hence, calculate the radius R of the helical path.

$$R = \dots\dots\dots \text{ m [2]}$$

[Total: 11]